

Continuous functions – 3

Level sets, discontinuities and the fine structure of the graph

1 сентября 2026 года

Theoretical minimum.

Theorem 1 (counting preimages). For a continuous $f: \mathbb{R} \rightarrow \mathbb{R}$ the number of solutions of $f(x) = y$ cannot be finite and even for all y ; parity obstructions of this kind follow from the intermediate value theorem applied on both sides of a local extremum.

Theorem 2 (one-sided limits). If f has one-sided limits at every point, then its set of discontinuities is at most countable. For a monotone f this set can be an arbitrary countable set — in particular, exactly $\mathbb{Q}$.

Fact 1. A continuous function on an interval has, at almost every point, a supporting line from below in the weak sense of Problem 4: this is the first step towards convexity arguments and towards the theory of shadow points in the homework.

1. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ have left and right limits at every point. Can it have uncountably many points of discontinuity? The same question for an arbitrary monotone function. Give an example of a monotone function discontinuous exactly at the rational points.
2. Does there exist a continuous $y = f(x)$ taking every value $y \in \mathbb{R}$
 - a) exactly three times;
 - b) exactly two times;
 - c) every value $y \in [0, 1]$ an even positive number of times;
 - d) every value $y \in [0, +\infty)$ an even positive number of times?
3. Does there exist a continuous $y = f(x)$ taking every value $y \in \mathbb{R}$ an even positive number of times?

4. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be continuous. Prove that for every open interval (a, b) there are $x_0 \in (a, b)$ and $l \in \mathbb{R}$ such that $f(x) \geq f(x_0) + l(x - x_0) + o(x - x_0)$ as $x \rightarrow x_0$.
5. (RMO 2024) Let $f, g: \mathbb{R} \rightarrow \mathbb{R}$ satisfy $g(x) = 2f(x) - f(2x)$ for all x .
- a) Prove that if f is bounded in a neighbourhood of the origin and g is continuous at the origin, then f is continuous at the origin.
 - b) Give an example of an f discontinuous at the origin for which g is continuous there.